

9. Scenario:

You work for a real estate agency and have been given a dataset containing information about properties for sale. The dataset is stored in a Pandas DataFrame named `property_data`. The DataFrame has columns for property ID, location, number of bedrooms, area in square feet, and listing price. Your task is to analyze the data and answer specific questions about the properties.

Question: Using Pandas DataFrame operations, how would you find the following information from the `property_data` DataFrame:

1. The average listing price of properties in each location.
2. The number of properties with more than four bedrooms.
3. The property with the largest area.

```
In [1]: import pandas as pd

property_data = pd.DataFrame({
    "property_id": [101, 102, 103, 104, 105, 106, 107, 108],
    "location": ["Chennai", "Coimbatore", "Chennai", "Madurai",
                 "Coimbatore", "Chennai", "Madurai", "Chennai"],
    "bedrooms": [3, 5, 2, 4, 6, 5, 3, 4],
    "area_sqft": [1500, 2500, 1200, 1800, 3200, 2800, 1700, 2100],
    "listing_price": [7500000, 9500000, 6000000, 7000000,
                     12000000, 11000000, 6800000, 8500000]
})

print(property_data)
```

	property_id	location	bedrooms	area_sqft	listing_price
0	101	Chennai	3	1500	7500000
1	102	Coimbatore	5	2500	9500000
2	103	Chennai	2	1200	6000000
3	104	Madurai	4	1800	7000000
4	105	Coimbatore	6	3200	12000000
5	106	Chennai	5	2800	11000000
6	107	Madurai	3	1700	6800000
7	108	Chennai	4	2100	8500000

1. Average listing price of properties in each location

```
In [2]: # 1. Average listing price of properties in each location
avg_price = property_data.groupby("location")["listing_price"].mean()
print("Average Listing Price by Location:")
print(avg_price)
```

```
Average Listing Price by Location:
location
Chennai      8250000.0
Coimbatore   10750000.0
Madurai      6900000.0
Name: listing_price, dtype: float64
```

2. Number of properties with more than four bedrooms

```
In [3]: count = property_data[property_data["bedrooms"] > 4].shape[0]
print("\nNumber of properties with more than 4 bedrooms:", count)
```

Number of properties with more than 4 bedrooms: 3

3. Property with the largest area

```
In [4]: largest = property_data.loc[property_data["area_sqft"].idxmax()]
print("\nProperty with the Largest Area:")
print(largest)
```

Property with the Largest Area:

property_id	105
location	Coimbatore
bedrooms	6
area_sqft	3200
listing_price	12000000

Name: 4, dtype: object

10. Scenario:

You are working on a data visualization project and need to create basic plots using Matplotlib. You have a dataset containing the monthly sales data for a company, including the month and corresponding sales values. Your task is to develop a Python program that generates line plots and bar plots to visualize the sales data.

Question:

1. How would you develop a Python program to create a line plot of the monthly sales data?
2. How would you develop a Python program to create a bar plot of the monthly sales data?

1. Python Program to Create a Line Plot

```
In [5]: import matplotlib.pyplot as plt

# Monthly sales data
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [25000, 32000, 28000, 40000, 37000, 45000]

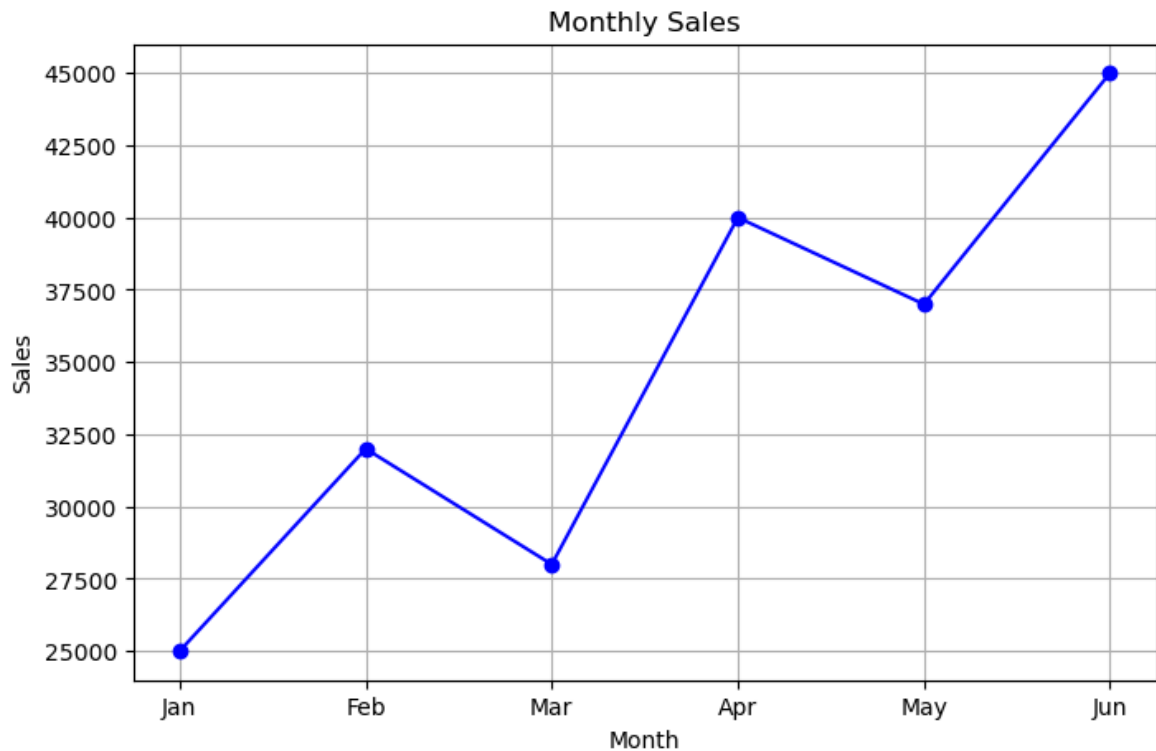
# Create Line plot
plt.figure(figsize=(8,5))
plt.plot(months, sales, marker='o', linestyle='--', color='blue')

# Add title and Labels
plt.title("Monthly Sales")
```

```
plt.xlabel("Month")
plt.ylabel("Sales")

# Display grid
plt.grid(True)

# Show plot
plt.show()
```



2. Python Program to Create a Bar Plot

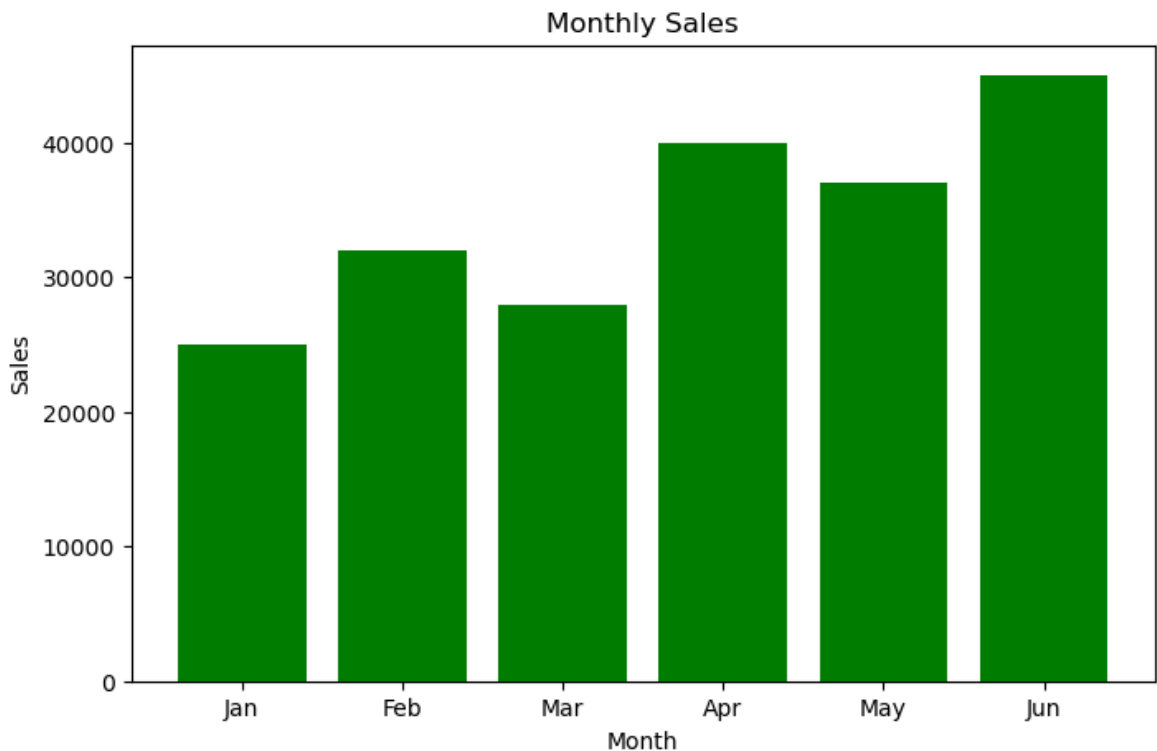
```
In [6]: import matplotlib.pyplot as plt

# Monthly sales data
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [25000, 32000, 28000, 40000, 37000, 45000]

# Create bar plot
plt.figure(figsize=(8,5))
plt.bar(months, sales, color="green")

# Add title and Labels
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")

# Show plot
plt.show()
```



11. Scenario :

You are a data scientist working for a company that sells products online. You have been tasked with creating a simple plot to show the sales of a product over time.

Question:

1. Write code to create a simple line plot in Python using Matplotlib to predict sales happened in a month?
2. Write code to create a scatter plot in Python using Matplotlib to predict sales happened in a month?
3. Develop a Python program to create a bar plot of the monthly sales data. n using Matplotlib to predict sales happened in a month?
4. Develop a Python program to create a bar plot of the monthly sales data.

1. Python Program to Create a Line Plot

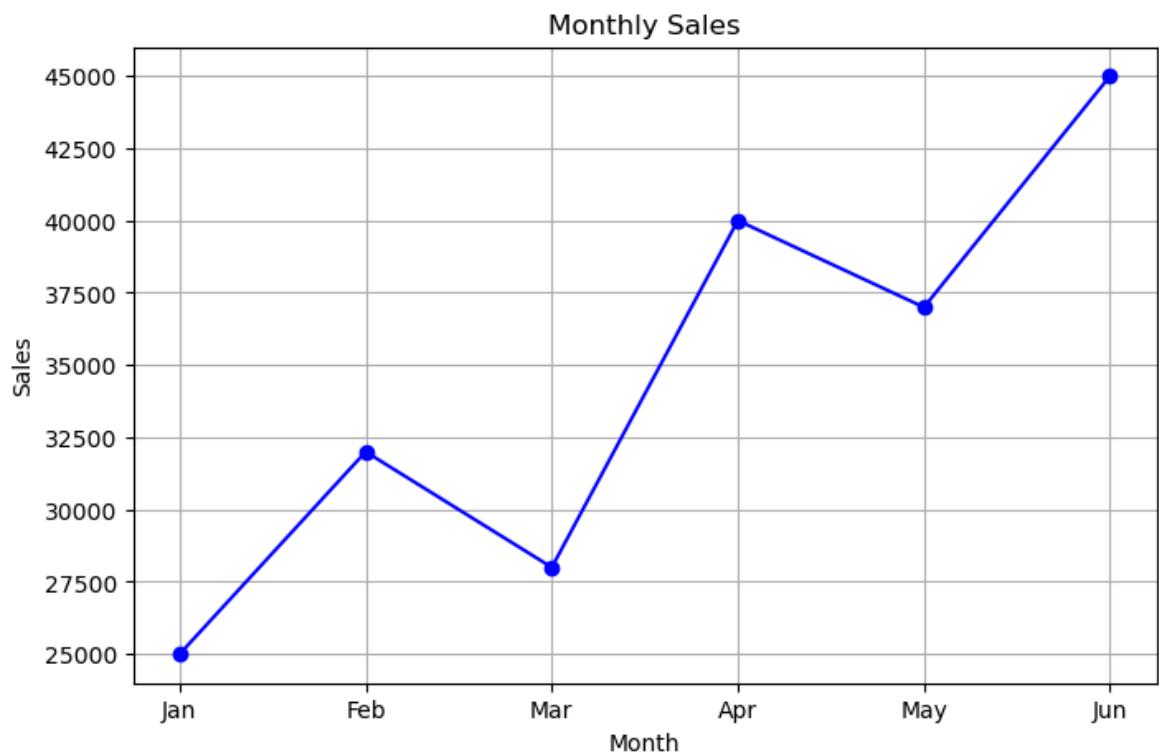
```
In [7]: import matplotlib.pyplot as plt

# Monthly sales data
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [25000, 32000, 28000, 40000, 37000, 45000]

# Line Plot
plt.figure(figsize=(8,5))
plt.plot(months, sales, marker='o', color='blue', linestyle='-')
```

```
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)

plt.show()
```



2. Python Program to Create a Scatter Plot

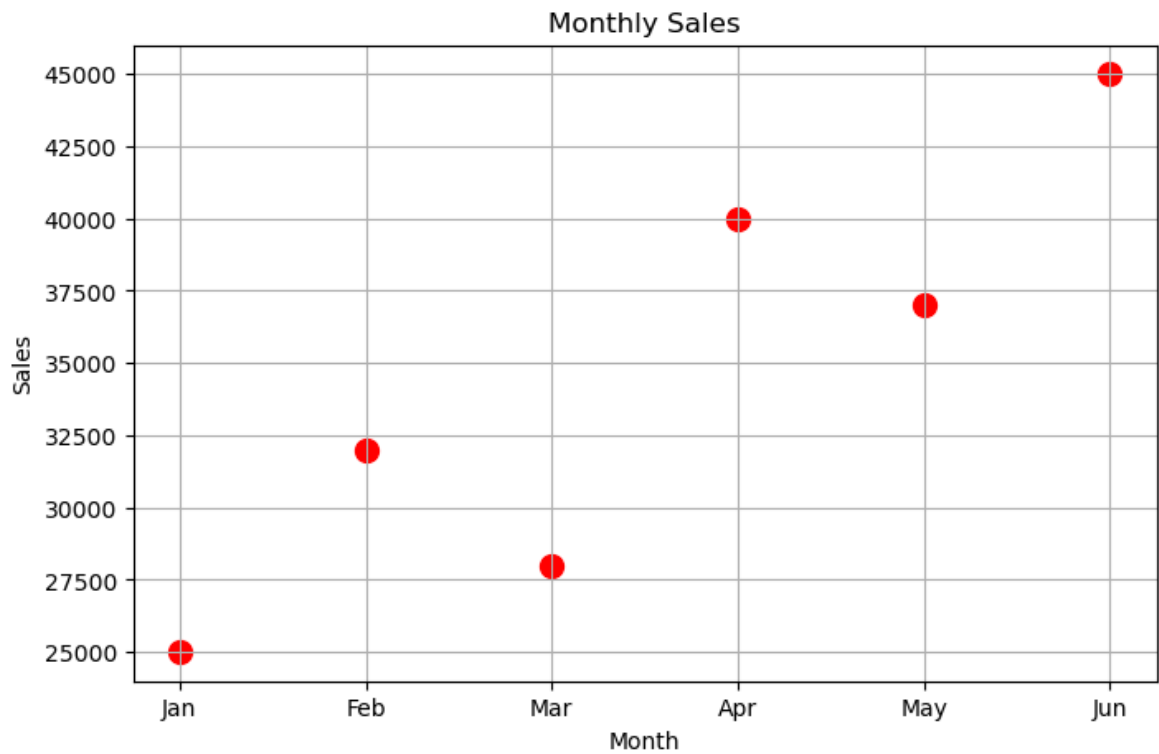
```
In [8]: import matplotlib.pyplot as plt

# Monthly sales data
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [25000, 32000, 28000, 40000, 37000, 45000]

# Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(months, sales, color='red', s=100)

plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)

plt.show()
```



3. Python Program to Create a Bar Plot

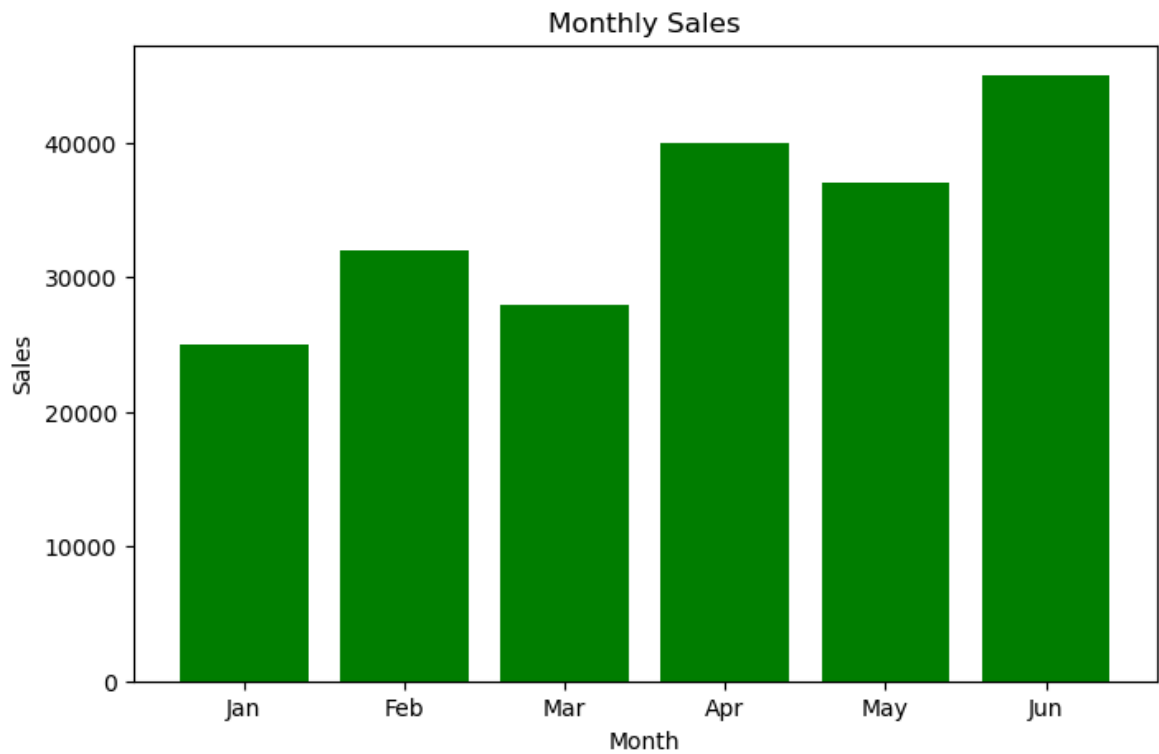
```
In [9]: import matplotlib.pyplot as plt

# Monthly sales data
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [25000, 32000, 28000, 40000, 37000, 45000]

# Bar Plot
plt.figure(figsize=(8,5))
plt.bar(months, sales, color='green')

plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



Line Plot

`plt.plot()` creates a line graph. Best for showing sales trends over time.

Scatter Plot

`plt.scatter()` creates a scatter graph. Displays individual sales data points.

Bar Plot

`plt.bar()` creates a bar chart. Best for comparing monthly sales values.

In []: